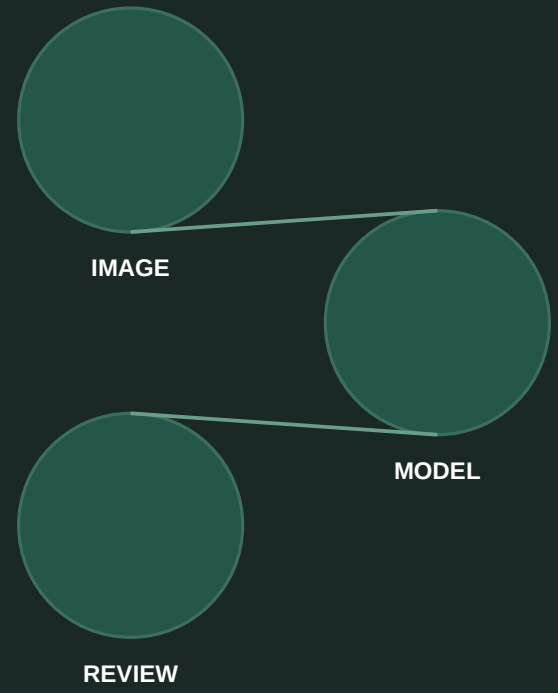
RESEARCH PROTOTYPE / PRESENTATION

A local-first brain MRI review and research classification workflow

What it does. How it was evaluated.
What its results do not prove.



A focused research workflow

The tool organizes image review and model experiments; it does not replace diagnostic interpretation.

INPUT

Single-frame MR DICOM, JPG, or PNG

>

PREPARE

Grayscale, denoise, resize, normalize

>

MODEL

Four-class research prediction

>

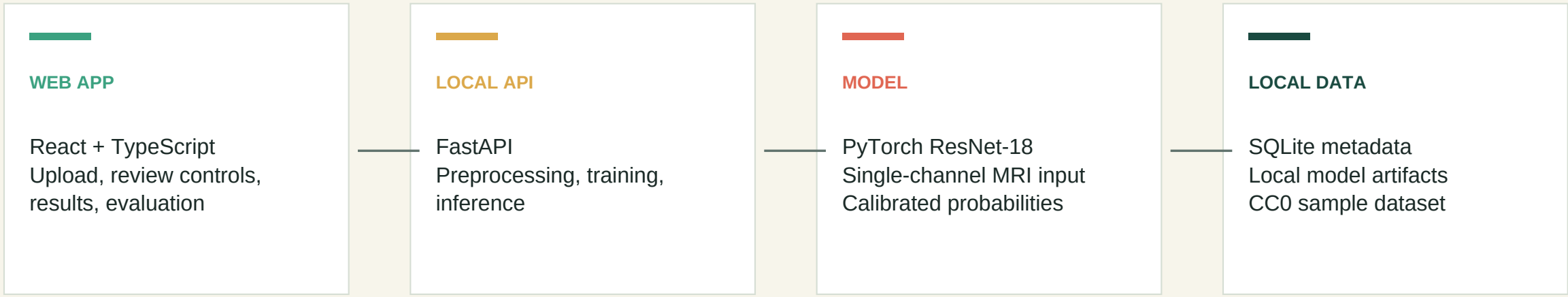
REVIEW

Probabilities and exploratory Grad-CAM

The model reports one of four dataset labels. It does not locate or segment a tumor, interpret a full MRI series, or establish a diagnosis.

Local-first architecture

Image bytes are processed locally; the database stores study metadata, not uploaded image files.



The service binds to loopback for local development. It has no authentication or multi-user access controls; do not expose it to a network without a security review.

From scan to review

A transparent processing path helps distinguish image handling from model interpretation.



Select

One 2D MRI image

Preprocess

Denoise + normalize

Infer

Class probabilities

Inspect

Output + Grad-CAM

Grad-CAM is an exploratory attention visualization - not a tumor boundary, segmentation, or clinical finding.

A labeled public sample dataset

The bundled CC0 dataset contains four image classes and separate source train/test folders.



7,023 images total | 5,969 source train | 1,054 source test | CC0-1.0

Source: Dharamanand/BrainTumorMRI on Hugging Face. Its card does not provide patient identifiers or document patient-level split construction.

How the current model was trained

ResNet-18 transfer learning, bounded local compute, and validation separated from final test scoring.

ARCHITECTURE	ImageNet-initialized ResNet-18; grayscale first layer; final block and classifier fine-tuned
TRAINING CAP	500 sampled images per class; 2,000 selected total
FIT / VALIDATION	1,600 fit images + 400 stratified validation images from the source training partition
AUGMENTATION	Training-only small rotation, translation, and scale changes
SELECTION	Best checkpoint chosen by validation loss; 10 epochs on CPU
FINAL TEST	All 1,054 source test images scored after checkpoint selection

This run used the source test split only after checkpoint selection; patient-level separation remains undocumented.

Held-out evaluation

Official source test partition; 1,054 images. Results are dataset-specific, not evidence of clinical accuracy.

ACCURACY	MACRO PRECISION	MACRO RECALL	MACRO F1
95.5%	**95.3%**	**95.4%**	**95.4%**
1,007 / 1,054 correct	Unweighted class average	Unweighted class average	Unweighted class average

CLASS-WISE RECALL

glioma	**95.9%**	support 219
meningioma	**91.6%**	support 261
no tumor	**98.4%**	support 306
pituitary	**95.9%**	support 268

WHAT THIS NUMBER MEANS

Performance on this source split only. Related slices may cross splits because patient-level separation is not documented.

Where predictions were confused

Rows are actual labels; columns are predicted labels. Diagonal cells are correct classifications.

ACTUAL CLASS \ PREDICTED CLASS	glioma	meningioma		pituitary
glioma	210	8	0	1
meningioma	11	239	4	7
no tumor	2	1	301	2
pituitary	2	9	0	257

READOUT

Meningioma has the lowest recall: 239 of 261 correct. Most errors in this class were predicted as glioma (11) or pituitary (7). This is a useful target for further dataset review.

Counts describe this test partition; they do not establish clinical sensitivity or generalization.

Calibration and explanations

Confidence is adjusted on validation data; Grad-CAM is exploratory and is not segmentation.

TEMPERATURE SCALING

1.29

validation-fitted temperature

VALIDATION NEGATIVE LOG-LIKELIHOOD



HELD-OUT EXPECTED CALIBRATION ERROR

1.45%

Measured on the source test split after temperature was fitted on the 400-image validation subset. ECE depends on the dataset and binning method.

Grad-CAM highlights model-sensitive regions; it does not prove why the model was right or wrong.

What the result does not establish

A strong score on one public image split is not the same as reliable performance in patient care.

PATIENT LEAKAGE

Patient identities and split construction are undocumented.

EXTERNAL VALIDITY

One source dataset; no multi-site or prospective evaluation.

TASK SCOPE

Four image labels; single 2D images only, not full MRI series.

CLINICAL READINESS

No clinical workflow validation, regulatory review, or deployment controls.

Not for diagnosis or treatment decisions. A qualified radiologist must review all findings.

A responsible path forward

Improve the evidence before increasing the claims.

01	**Source stronger data**	Use institution-approved, patient-identified splits and verified labels.
02	**Lock external evaluation**	Keep an independent site or cohort untouched until final reporting.
03	**Measure clinical risks**	Review class-wise sensitivity, calibration, scanner shift, and failure cases.
04	**Require governance**	Add clinical oversight, privacy controls, auditability, and regulatory review before use.

Current takeaway: a working local research prototype with promising source-split results and important validation gaps.

Opening and workflow

Suggested plain-language narration; adapt to your audience.

Slide 1 - Neuro Imaging Lab

Introduce this as a research prototype for exploring a local MRI review workflow. It is not a medical device and does not make clinical decisions.

Slide 2 - A focused research workflow

Walk through the four stages from a single image to a reviewable model output. Emphasize that the prototype classifies an image into one of four labels and does not perform tumor segmentation or radiology interpretation.

Slide 3 - Local-first architecture

Explain that the browser, API, model, and data artifacts run on the user?s machine. Uploaded images are handled in memory, while case history is limited to filenames and timestamps. The current development service should remain local.

Slide 4 - From scan to review

Describe the user path: select a supported image, preprocess it, run inference, and review probabilities. Grad-CAM highlights image regions that influenced the network but is not a tumor mask.

Data and results

Suggested plain-language narration; adapt to your audience.

Slide 5 - Labeled public sample dataset

The app uses the 7,023-image public dataset shown here. The license is listed as CC0-1.0. Explain that class labels and image-level splits are available, but the source does not establish that patients are separated across splits.

Slide 6 - How the current model was trained

The training cap selected up to 500 images per class, then the selected images were split into 1,600 fitting and 400 validation examples. The test partition was not used to select epochs in this run. The older compact-CNN result is not a fair baseline because it was selected against test loss.

Slide 7 - Held-out evaluation

Read the aggregate metrics, then call out the lowest class recall: meningioma at 91.6 percent. This is measured on the source test partition. Because patient separation is unknown, the result may overestimate performance on new patients or scanners.

Risks and close

Suggested plain-language narration; adapt to your audience.

Slide 8 - Where predictions were confused

Explain how to read the confusion matrix: rows are actual class, columns predicted. The diagonal is correct. Meningioma has the highest number of missed cases in this split and should be prioritized for independent review.

Slide 9 - Calibration and explanations

Temperature scaling adjusts the confidence distribution without changing the predicted class. It reduced validation NLL from 0.176 to 0.167. The 1.45 percent test ECE is a dataset-specific calibration measure, not a guarantee. Grad-CAM is only an exploratory visualization.

Slide 10 - What the result does not establish

Be direct: the data lacks documented patient-level separation, so related slices could appear across train and test. The sample is from one source and does not represent every scanner, population, or protocol. The app is for research demonstration only.

Slide 11 - A responsible path forward

Close with the work required before any real-world decision support: patient-level labels, an untouched external cohort, scanner-shift tests, clinical governance, and review by qualified professionals. The current result is promising for a demo, not a clinical claim.